American Computer Science League

2020-2021 ● Contest 3: Solutions ● Classroom Division

1. Boolean Algebra

$$A(\overline{B+C}) + A\overline{B}C = A\overline{B}\,\overline{C} + A\overline{B}C$$
$$= A\overline{B}(\overline{C} + C)$$
$$= A\overline{B}$$

1. $A\overline{B}$ (B)

2. Boolean Algebra

$$A\,(\overline{\overline{A+B}})(\overline{AB}) = A(\overline{\overline{A+B}}) + \overline{\overline{AB}}$$
$$= AA + AB + AB$$
$$= A + AB$$
$$= A(1 + B)$$
$$= A$$

Therefore there are 2 ordered pairs that make the expression TRUE:
(1, 1) and (1, 0) or (1, *)

2. 2 (C)

3. Boolean Algebra

$$A(\overline{B} + \overline{C}) + ABC = A\overline{B} + A\overline{C} + ABC$$
$$= A(\overline{B} + \overline{C} + BC)$$

To be TRUE, $A = 1$ and $\overline{B} + \overline{C} + BC = 1$.
If $B = 1$, then $0 + \overline{C} + C = 1 \Rightarrow C = *$
If $B = 0$, then $1 + \overline{C} + C = 1 \Rightarrow C = *$
Therefore (1, 1, *) and (1, 0, *) satisfy the expression or (1, *, *).

3. (1, *, *) (C)

4. Boolean Algebra

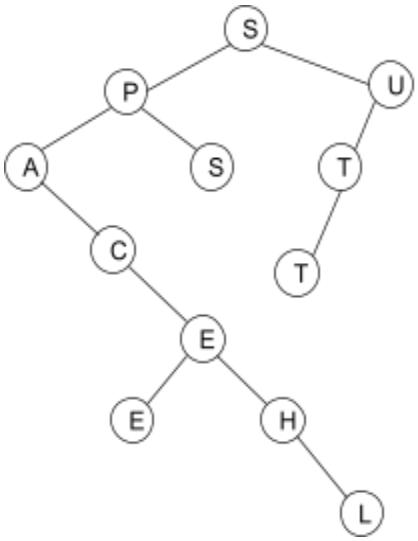
$$\overline{B}\,(\overline{A + C}) + (\overline{B\,C}) = \overline{B}\,\overline{A}\,\overline{C} + \overline{B} + \overline{C}$$
$$= \overline{B}\,(\overline{A}\,\overline{C} + 1) + \overline{C}$$
$$= \overline{B} + \overline{C}$$

To be TRUE, $\overline{B} + \overline{C} = 1$ which is TRUE unless $B = 1$ and $C = 1$.
Therefore, only (0,1,1) and (1,1,1) make it FALSE so 6 ordered triples make the expression TRUE.

4. 6 (B)

5. Data Structures

The binary search tree for SPACESHUTTLE is:



There are 5 nodes with only one child: U, A, T, C, H

5. 5 (D)

6. Data Structures

The stack is constructed using LIFO as follows:

PUSH(H), PUSH(A), POP(X), PUSH(N), PUSH(S), POP(X), PUSH(S), POP(X), PUSH(O), PUSH(L), POP(X), PUSH(O), POP(X), POP(X)

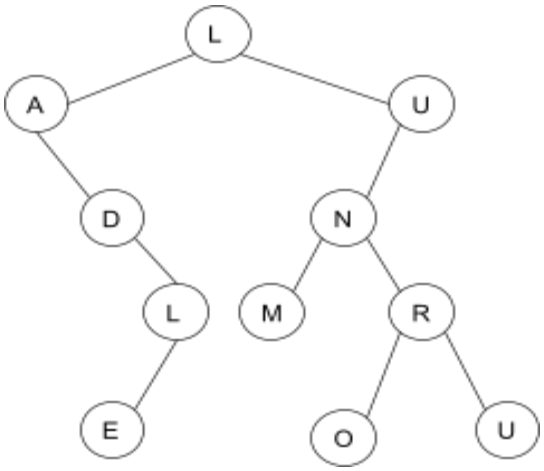
H, HA, H, HN, HNS, HN, HNS, HN, HNO, HNOL, HNO, HNOO, HNO, HN.

The next item popped would be an N.

6. N (E)

7. Data Structures

The binary search tree for LUNARMODULE is:



The depth of this tree is 4.

7. 4 (D)

8. Data Structures

The queue is constructed using FIFO as follows:

D, DA, DAR, AR, ART, ARTH, RTH, RTHV, RTHVA, RTHVAD, THVAD, HVAD, HVADE, HVADER, VADER, ADER, DER

The next item popped is D.

8. D (B)

9. FSAs & Regular Expressions

Given the regular expression: `0(0*1U1*0)1*0*1`

A. 000010 - must end with a 1
B. 0101010101 - must have 3 or fewer sets of 0s and 1s
C. 0000001 - 000*1
D. 01110110001 - 01*01*0*1
E. 11111111 - must begin with a 0
F. 0001111 - 00*11*1
G. 0101 - 010*1
H. 00001110001 - 00*11*0*1

9. A, B, E (A)

10. FSAs & Regular Expressions

The Regular Expression is: **`[^s][aeiou][p-t]*(s|er)`**

A. tater - valid
B. sorts - string cannot begin with an s
C. faster - valid
D. plaques - string cannot start with 2 non-vowels
E. deer - valid
F. rooster - second o is not in p-t range for third letter
G. hits - valid
H. bursts - valid

10. B, D, F (E)